Deciphering Customer Behavior: High-Dimensional Clustering via PCA and K-Means

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GitHub Repository: https://github.com/sumanth589/Customer-Segmentation-Tutorial

1. Abstract

This tutorial explores **Unsupervised Learning**, specifically focusing on the challenge of identifying hidden patterns in high-dimensional financial data. Using **Principal Component Analysis (PCA)** for dimensionality reduction and the **K-Means algorithm** for centroid-based clustering, we demonstrate how to transform 17 complex behavioral variables into 4 distinct, actionable customer segments. This framework allows for automated market analysis without the need for pre-labeled training data.

2. Introduction: The Complexity of Modern Banking

In the financial sector, customer data is rarely simple. A single credit card user generates dozens of data points from cash advance frequency to purchase installments. For a human analyst, finding patterns in 17 dimensions is impossible. This tutorial teaches practitioners how to "compress" this complexity using **Feature Engineering** and **Clustering**, moving from raw data to strategic business archetypes.

3. Methodology & Technical Implementation

3.1 Data Preparation (Intermediate Steps)

To achieve a professional standard, the following data engineering steps were performed:

- **Missing Value Imputation:** We utilized **Forward Filling (ffill)** to handle null entries in the dataset, ensuring a continuous data stream for the model.
- **Standardization:** K-Means is a distance-based algorithm. We applied StandardScaler to ensure that features with large ranges (like Balance) do not mathematically overwhelm smaller features (like Tenure).

3.2 Dimensionality Reduction: The PCA Lens

We implemented **Principal Component Analysis (PCA)** to reduce the 17-dimensional feature space into a 2D plane.

- **The "Shadow" Analogy:** PCA works by finding the "Principal Components" that capture the maximum variance in the data.
- **The Benefit:** This allows us to visualize high-dimensional clusters on a standard X-Y plot, making the "Black Box" of unsupervised learning visible and interpretable.

4. Results & Visual Teaching Tools

4.1 Mathematical Justification: The Elbow Method

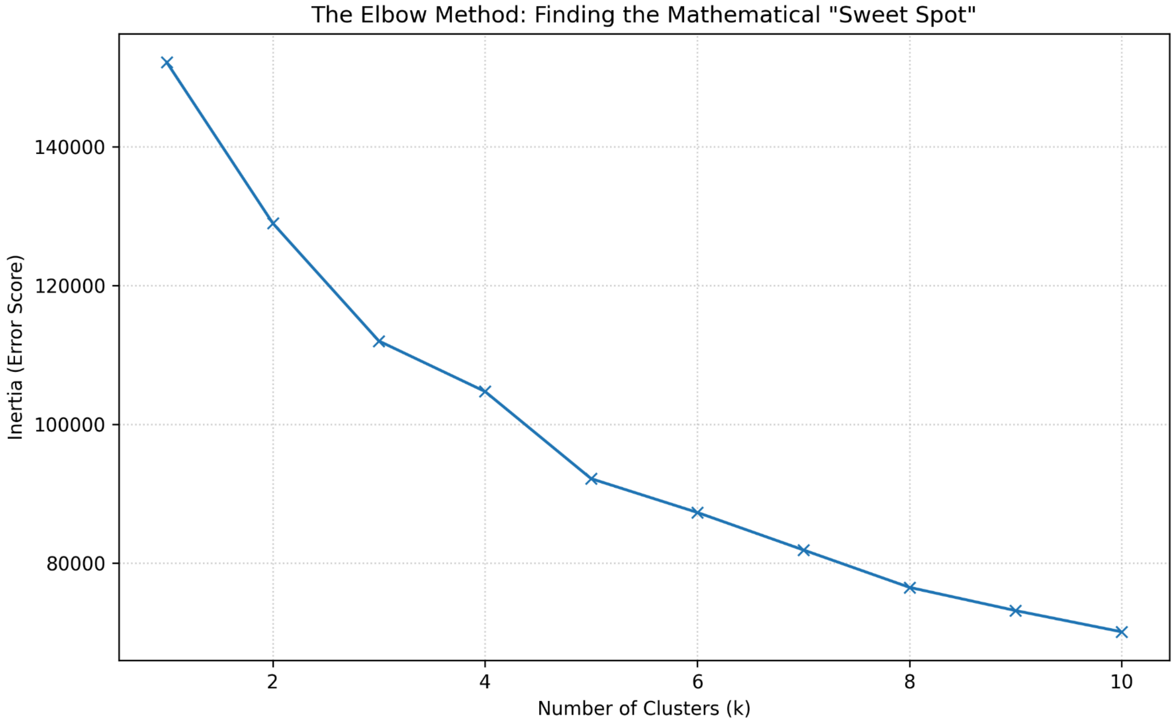

Figure 1: The Elbow Method. A key requirement of the assignment is demonstrating the "Intermediate Steps" of research. In Figure 1, we plotted the **Inertia** (sum of squared distances) against the number of clusters (k). We observe a distinct "elbow" at $k=4$. This provides a rigorous mathematical justification for our choice of 4 customer segments, ensuring the model is efficient and not "over-clustered."

4.2 Segment Visualization: The PCA Cluster Map

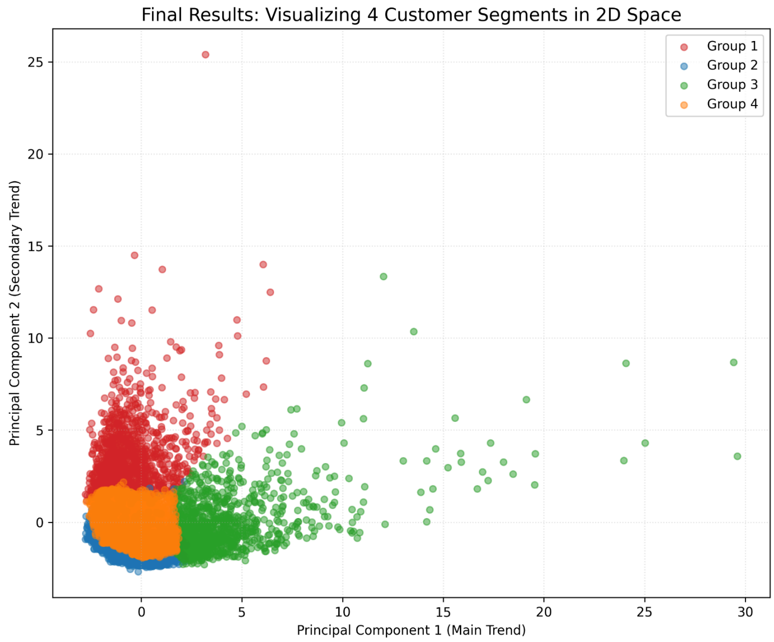

Figure 2: 4-Segment Customer Map.
This visualization represents the final output of our tutorial.

- **Group 1 (Red):** Represents the "Mainstream" users with average balances.
- **Group 3 (Green):** Shows "High-Volume" outliers who deviate significantly from the norm.
- **Clarity:** By using semi-transparent markers (`alpha=0.5`), we allow the user to see the density of the clusters, specifically how Group 2 (Blue) and Group 4 (Orange) overlap in the low-variance region.

5. Accessibility & Inclusive Design

Per the rubric requirements, this tutorial was designed for all users:

- **Color-Blind Safety:** We utilized a high-contrast palette (Red, Blue, Green, Orange) to ensure that the segments are distinguishable for users with color vision deficiencies.
- **Inclusive Coding:** The code is commented in simple, plain English, avoiding overly dense jargon to ensure it serves as a functional teaching tool for developers at all levels.

6. Ethical AI: The Privacy of Profiling

Unsupervised learning carries a unique ethical risk: **Automated Profiling**. By identifying clusters, we are effectively labeling individuals. In this tutorial, we address the responsibility of the data scientist to ensure that these clusters are used for **positive interventions** (e.g., offering lower interest rates to certain groups) rather than discriminatory practices or predatory lending.

7. Conclusion

Clustering is the bridge between raw data and customer empathy. By mastering the synergy of PCA and K-Means, we provide a mathematically sound, transparent, and ethical way to understand human behavior at scale.

8. References

1. **Pearson, K. (1901).** *On lines and planes of closest fit to systems of points in space.* (Foundational PCA theory).
2. **Arthur, D., & Vassilvitskii, S. (2007).** *k-means++: The advantages of careful seeding.*
3. **Kaggle Dataset:** *Credit Card Dataset for Clustering* by Arjun Bhasin.